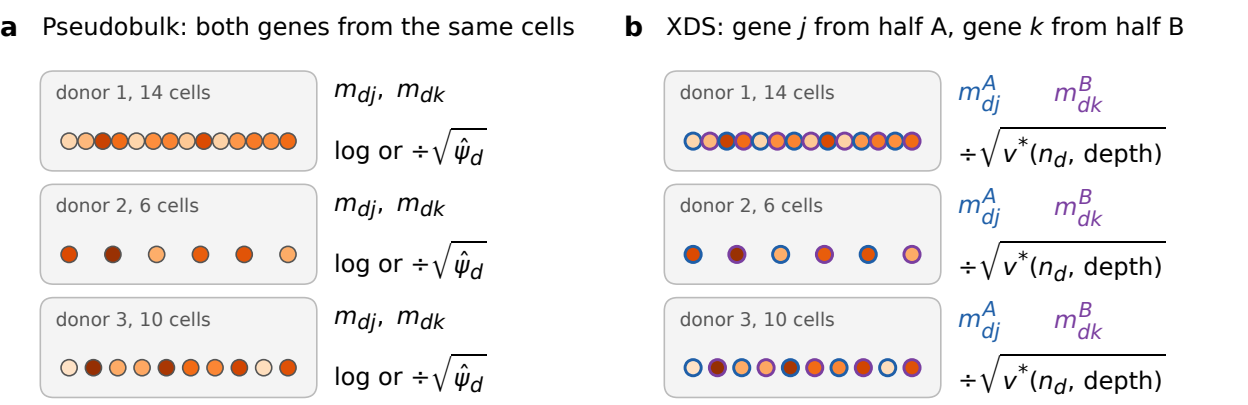
a
Pseudobulk: both genes from the same cells
donor 1, 14 cells
$m_{dj},\ m_{dk}$
log or $\div\sqrt{\hat{\psi}_d}$
donor 2, 6 cells
$m_{dj},\ m_{dk}$
log or $\div\sqrt{\hat{\psi}_d}$
donor 3, 10 cells
$m_{dj},\ m_{dk}$
log or $\div\sqrt{\hat{\psi}_d}$
cell-state programs leak in
(fraction $\kappa_d \approx 1/n_d$); the transform's
bias depends on n_d, shared by all genes
b
XDS: gene j from half A, gene k from half B
donor 1, 14 cells
m^A_{dj} m^B_{dk}
$\div\sqrt{v^*(n_d,\ \text{depth})}$
donor 2, 6 cells
m^A_{dj} m^B_{dk}
$\div\sqrt{v^*(n_d,\ \text{depth})}$
donor 3, 10 cells
m^A_{dj} m^B_{dk}
$\div\sqrt{v^*(n_d,\ \text{depth})}$
no shared cells: no leak; design-only scale:
no n_d-dependent bias; test
$\sum_d R^A_{dj} R^B_{dk}$ by donor permutation